

# Binary Batch Distillation

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                   | Points | Grading criteria                                                                                                                                                                                                                                                                                                                            | Score |
|---------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5      | State the 10 wt% ethanol/water system, eight-tray batch column, and objectives of both experiments: boil-up, pressure drop, foaming and efficiency at total reflux; composition versus time at different reflux ratios.                                                                                                                     | _____ |
| Apparatus                 | 5      | Introduce a numbered, captioned figure and matching component table. Identify the reboiler, trays, condenser/cooling-water system, reflux controller, receivers, sampling valves V2/V3, manometer valves V6/V7, temperature points and console. Describe component functions accurately.                                                    | _____ |
| Materials and Supplies    | 5      | List the 10-L charge, ethanol, DI feed water and tap cooling water separately. Include preparation vessels/funnel, graduated cylinders, timer, sample tubes/caps, sample-cooling supplies and densitometer/Snap 41, with quantities and purposes.                                                                                           | _____ |
| Experimental Plan         | 10     | Specify setup, warmup, stabilization and shutdown. For A, use 0.75, 1.00 and 1.25 kW; record repeated V3 boil-up, pressure drops, foaming, temperatures and V2/V3 compositions. Open and close V6 before V7. For B, specify reflux ratios and timed samples. Include sample cooling, explicit variables and controlled initial composition. | _____ |
| Project Safety Evaluation | 10     | Address ethanol/ignition control, ventilation, hot samples/surfaces, electrical hazards and vent blockage. State operating conditions, glove type/PPE, low-level interlock, circuit breakers, shutoffs and spill response. Complete SDS/CAS information, safety acknowledgment, signatures and dates.                                       | _____ |
| Content total             | 35     |                                                                                                                                                                                                                                                                                                                                             | _____ |

### Late-submission policy

By the due date: full earned score.

After the due date and up to the next lab period: 50% of the earned score.

After the next lab period: 0 score.

Content score: \_\_\_\_\_ / 35    Late multiplier: 1 / 0.5 / 0    Final score: \_\_\_\_\_ / 35

# Evaporation

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                   | Points | Grading criteria                                                                                                                                                                                                                                                                                                                                                                                                   | Score |
|---------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5      | Determine capacity and steam economy of a single-effect evaporator containing 10 wt% sugar solution as vacuum/absolute pressure changes, while maintaining solution level and concentration.                                                                                                                                                                                                                       | _____ |
| Apparatus                 | 5      | Introduce a numbered, captioned figure and matching component table. Identify the separator/recirculation loop, steam heat exchanger, condenser, steam trap, condensate/product flasks, vacuum pump/bypass/gauge, makeup tank/valve, cooling-water measurements and thermocouples/reader.                                                                                                                          | _____ |
| Materials and Supplies    | 5      | List sugar and DI water for 10 wt% solution, DI makeup water, steam and cooling water. Include balance, mixing vessel, scoop/funnel, graduated cylinder, timer and collection flasks, with quantities and purposes.                                                                                                                                                                                                | _____ |
| Experimental Plan         | 10     | Drain, prepare/fill the solution and makeup tank, and establish steam, cooling water and vacuum. Maintain level and concentration. Use three vacuum settings and a defined temperature-stability criterion. Record pressures, temperatures, cooling-water data and timed product/steam-condensate volumes, including a second reading after 5–10 minutes. Explicitly identify variables and controlled conditions. | _____ |
| Project Safety Evaluation | 10     | Address steam, hot surfaces/condensate/flasks, vacuum/pressure, pump and electrical-water hazards. State pressures and temperatures with correct units, thermal gloves/PPE, shutoffs, pump switch/bypass and protective shield. Complete SDS/CAS information, safety acknowledgment, signatures and dates.                                                                                                         | _____ |
| Content total             | 35     |                                                                                                                                                                                                                                                                                                                                                                                                                    | _____ |

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# Fluid Flow in Pipe Systems

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                   | Points | Grading criteria                                                                                                                                                                                                                                                                                                                                                                    | Score |
|---------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5      | Determine orifice flow coefficient $C_f$ , pipe friction factor $f$ and fitting loss coefficient $K$ from water-flow, pressure-drop and temperature measurements at varied flows. Identify the H-6920 apparatus and comparisons with Moody/Colebrook and relevant fitting/orifice correlations.                                                                                     | _____ |
| Apparatus                 | 5      | Introduce a numbered, captioned figure and matching component table. Identify pump, reservoir, rotameter, nine copper/plastic/steel pipe branches, flow/shutoff valves, orifice, fittings, pressure taps, differential-pressure transducers/gauges and temperature measurement. Describe functions accurately.                                                                      | _____ |
| Materials and Supplies    | 5      | List water, large collection bucket, graduated vessel, stopwatch/timer, pressure tubing and quick-connects. Provide relevant pipe/fitting specifications and the purpose of each measurement or calibration supply.                                                                                                                                                                 | _____ |
| Experimental Plan         | 10     | Include timed-volume rotameter calibration, orifice testing, all nine straight-pipe branches and fitting studies, including the right-angle bend. Use at least three flow rates per study. Record $Q$ , pressure drop, temperature and geometry/tap spacing. Specify pressure connections, stabilization, safe reconnection/shutdown, and explicit independent/dependent variables. | _____ |
| Project Safety Evaluation | 10     | Address pump/moving parts, dry running, blocked flow, pressure release before reconnection, water hammer, leaks, slips and electrical-water hazards. State pressure limits, PPE, pump switch, shutoffs and circuit breaker. Complete SDS/CAS information, safety acknowledgment, signatures and dates.                                                                              | _____ |
| Content total             | 35     |                                                                                                                                                                                                                                                                                                                                                                                     | _____ |

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# Humidification

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                   | Points    | Grading criteria                                                                                                                                                                                                                                                                                                                                                                                                        | Score |
|---------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5         | Determine heat- and gas-phase mass-transfer coefficients in the wetted-wall column from humidity, temperature and flow measurements; compare with correlations. Specify three air flow rates and three inlet-air temperatures, with assigned water-flow and air-pressure conditions controlled.                                                                                                                         | _____ |
| Apparatus                 | 5         | Introduce a numbered, captioned figure and matching component table. Identify the column/weir, reservoir, pump, water valve/rotameter, air regulator/gauge/valve/rotameter, steam-air heat exchanger, inlet/internal/interface/water temperature measurements, reader and outlet humidity measurement.                                                                                                                  | _____ |
| Materials and Supplies    | 5         | List DI water, compressed air and steam separately. Include hygrometer, bucket/beaker, filling funnel, temperature-measurement supplies and stepladder if used. State useful quantities, specifications and purposes.                                                                                                                                                                                                   | _____ |
| Experimental Plan         | 10        | Record baseline inlet temperature/humidity; fill, circulate water and establish full wall wetting. Use three air flows at ambient and two heated inlet conditions. Specify target settings, steady state and controlled water-flow/pressure conditions. Record required flow, humidity and inlet/outlet/internal/interface temperatures. Explicitly distinguish varied, controlled, measured and calculated quantities. | _____ |
| Project Safety Evaluation | 10        | Address electrical-water contact, pump dry running/moving parts, compressed-air pressure/hoses, hot steam/exchanger, ladder use and wet floors. State source/operating pressures, temperatures, thermal gloves/PPE, air/steam shutoffs and pump switch. Complete SDS/CAS information, safety acknowledgment, signatures and dates.                                                                                      | _____ |
| <b>Content total</b>      | <b>35</b> |                                                                                                                                                                                                                                                                                                                                                                                                                         | _____ |

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# Membrane Air Separation

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                   | Points    | Grading criteria                                                                                                                                                                                                                                                                                                                                                    | Score |
|---------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5         | Identify the two-module hollow-fiber oxygen/nitrogen separation system. Compare series and parallel operation as tube-side pressure and flow vary; measure outlet flows/compositions and determine transport coefficients and separation performance.                                                                                                               | _____ |
| Apparatus                 | 5         | Introduce a numbered, captioned figure and matching component table of installed equipment. Identify modules, series/parallel flow paths, valves A–D, pressure sensing, tube/shell flow meters, oxygen analyzers, vents and operating controls. Describe functions accurately.                                                                                      | _____ |
| Materials and Supplies    | 5         | List the dry compressed-air cylinder, regulator, hoses/fittings and required instrument power/batteries or oxygen-sensor support supplies. State relevant specifications, ratings and purposes.                                                                                                                                                                     | _____ |
| Experimental Plan         | 10        | Warm/check analyzers with room air and record ambient temperature. Set parallel A-left/C-down or series A-right/C-up; B/D up to sample and down to vent, never middle. Specify pressure/flow matrix, stabilization, tube/shell flow and oxygen measurements, safe cylinder operation/shutdown, and explicit varied, controlled, measured and calculated quantities. | _____ |
| Project Safety Evaluation | 10        | Address secured high-pressure cylinders, regulator/component limits, pressure relief, hose/jet hazards and atmospheric shell venting. Include correct valve positions, grounding/GFCI, eye protection, depressurization and emergency shutoff. Complete operating conditions, SDS/CAS information, safety acknowledgment, signatures and dates.                     | _____ |
| <b>Content total</b>      | <b>35</b> |                                                                                                                                                                                                                                                                                                                                                                     | _____ |

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# Non-Newtonian Fluid

## Pre-Lab Rubric • 35 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

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| Section                   | Points | Grading criteria                                                                                                                                                                                                                                                                                                                                                                                                           | Score |
|---------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Experimental Objective    | 5      | Determine m and n for approximately 1 wt% Carbopol 934/940 using the capillary viscometer. State pressure/flow measurements, variation in tube length/diameter and pressure, and the fluid-classification/geometry objective.                                                                                                                                                                                              | _____ |
| Apparatus                 | 5      | Introduce a numbered, captioned figure and matching component table. Identify regulator, gauge, vessel, air-line shutoff, pressure-relief/vent and outlet valves, and capillary connection. Describe functions correctly; list the mixer, scale and other supporting tools under Materials/Supplies.                                                                                                                       | _____ |
| Materials and Supplies    | 5      | List Carbopol grade, powder mass and approximately 2 L of 1 wt% solution; water; 1 N NaOH and amount; compressed air; mixer/rheostat, lab jack, scale, spatula, preparation/storage/collection vessels, graduated cylinder, pH paper and timer. State purposes and concentrations explicitly.                                                                                                                              | _____ |
| Experimental Plan         | 10     | Give consistent preparation quantities and neutralization to pH 5–7. Secure the vessel; keep vent open while filling, close last to pressurize and open first to depressurize. Isolate air and verify zero pressure before tube changes. Test identified geometries at 3–4 pressures, $\leq 60$ psig; obtain Q from timed volume or mass/density. Explicitly state variables, controlled concentration and calculated m/n. | _____ |
| Project Safety Evaluation | 10     | Distinguish 120-psig facility air from the 60-psig operating limit. Address overpressure, safe tube changes, NaOH corrosivity, Carbopol dust, rotating mixer and electrical-water hazards. Identify shutoff/vent/relief controls, glove type and PPE. Complete SDS/CAS information, safety acknowledgment, signatures and dates.                                                                                           | _____ |
| Content total             | 35     |                                                                                                                                                                                                                                                                                                                                                                                                                            | _____ |

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# Binary Batch Distillation

## Final Report Rubric • 100 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                                                                                                            | Score |
|----------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | State both objectives, feed, apparatus, method and operating conditions. Give numerical results for Experiments A and B and supported conclusions about separation and operating performance.                                                                                                                                                                                               | _____ |
| Introduction               | 10         | Describe engineering significance, applications, the batch ethanol–water system and apparatus. State the scope of Experiments A and B and their variables. Cite background information.                                                                                                                                                                                                     | _____ |
| Theory                     | 10         | Present vapor–liquid equilibrium, relative volatility, Fenske stages and efficiency, boil-up, pressure drop, foaming, reflux and batch McCabe–Thiele analysis. Define symbols, units and assumptions; number and cite equations. Include the energy balance if reported or assigned.                                                                                                        | _____ |
| Results                    | 15         | A: boil-up; top/bottom/overall pressure drops; foaming; temperature–time profiles; compositions; Fenske stages/efficiency; log-log pressure-drop–boil-up and efficiency–boil-up plots. Ending at 1.25 kW is acceptable. B: composition–time plots at $\geq 2$ reflux ratios; McCabe–Thiele comparison. Give units, captions and conditions; energy-balance outputs if reported or assigned. | _____ |
| Discussion                 | 15         | Quantitatively explain Experiment A trends, operating limits, efficiency and foaming, and Experiment B reflux–composition trends. Compare results with theory, McCabe–Thiele analysis and industry evidence. Address anomalies and support conclusions. Discuss the energy balance if reported or assigned.                                                                                 | _____ |
| References                 | 10         | Use credible sources and cite outside claims in the text. Provide complete, consistent entries and cite every listed source. Meet the source-count requirement when assigned.                                                                                                                                                                                                               | _____ |
| Appendix I: Data/Graphs    | 10         | Include complete raw data for both experiments, with composition, boil-up, pressure, temperature, time, reflux and operating conditions. Provide traceable supporting tables and plots with units and readable captions.                                                                                                                                                                    | _____ |
| Appendix II: Error         | 10         | Identify specific random, systematic and model errors. Evaluate repeatability and uncertainty or sensitivity. Quantify effects on composition, stages, efficiency, trends or conclusions, and propose realistic improvements.                                                                                                                                                               | _____ |
| Appendix III: Calculations | 10         | Identify the run. Show boil-up, composition conversion, relative volatility, Fenske stages, efficiency, pressure and McCabe–Thiele/reflux calculations, with substitutions and units. Include energy-balance calculations if reported or assigned. Match calculated values to reported results.                                                                                             | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                                                                                                             | _____ |

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# Evaporation

## Final Report Rubric • 100 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

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| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                                                | Score |
|----------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | Identify the evaporator, fluid, objective, method and actual vacuum range. Report numerical capacity, steam economy and boiling-temperature/pressure results, the main trend and a supported conclusion.                                                                                                                        | _____ |
| Introduction               | 10         | Describe industrial relevance and the single-effect process and apparatus. State the objective, scope, varied and controlled variables, and planned analyses. Cite relevant background information.                                                                                                                             | _____ |
| Theory                     | 10         | Present vacuum, boiling-point and temperature-driving-force relationships; $q = U A \Delta T$ ; mass/energy balances; and vacuum-to-absolute-pressure conversion. Define capacity as evaporated-water mass per time and economy as evaporated-water mass per steam mass. Define symbols, units and assumptions; cite equations. | _____ |
| Results                    | 15         | Present capacity and steam economy versus absolute pressure in the main text, with numerical values for every condition. Include boiling temperature, $\Delta T$ and applicable energy-balance results. Label conditions, units, tables and figures clearly.                                                                    | _____ |
| Discussion                 | 15         | Quantitatively connect pressure, boiling temperature, $\Delta T$ , heat transfer, capacity and steam economy. Evaluate single-effect economy, applicable energy-balance results and limitations, theory and literature comparisons, assumptions, deviations and improvements.                                                   | _____ |
| References                 | 10         | Meet the assigned source-count requirement. Cite all listed sources in the text, number citations in first-use order, and provide complete entries that allow each source to be retrieved.                                                                                                                                      | _____ |
| Appendix I: Data/Graphs    | 10         | Include all trials: pressures, temperatures, cooling-water data, product and steam-condensate volumes and times, condition identifiers and constant-level operation. Provide traceable capacity, economy and applicable energy calculations, units and supplemental plots.                                                      | _____ |
| Appendix II: Error         | 10         | Evaluate pressure, volume/time, temperature/flow, level and heat-loss errors qualitatively and quantitatively. Report uncertainty or percent difference, relate effects to capacity, economy and applicable energy results, and propose improvements.                                                                           | _____ |
| Appendix III: Calculations | 10         | Identify a trial. Show pressure conversion, product and steam rates, capacity, economy and applicable energy-balance calculations. Include equations, substitutions, intermediate units and results matching the reported tables.                                                                                               | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                                                 | _____ |

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# Fluid Flow in Pipe Systems

## Final Report Rubric • 100 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                                          | Score |
|----------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | State the purpose, four studies, methods and operating conditions. Report numerical calibration results and representative $C_f$ , $f$ and $K$ values. Compare with theory and give supported conclusions.                                                                                                                | _____ |
| Introduction               | 10         | Describe engineering relevance and the H-6920 water-flow system. State all four objectives, pipe and fitting scope, and measured variables. Support the background with citations.                                                                                                                                        | _____ |
| Theory                     | 10         | Present continuity/Bernoulli equations, orifice $C_f$ , Reynolds number, Darcy and Fanning friction factors, head loss, roughness, Colebrook/Moody relationships, and fitting $K$ /equivalent length. Define quantities, units, applicable flow regimes and assumptions; cite models.                                     | _____ |
| Results                    | 15         | Report the rotameter calibration fit/conversion; orifice-only $C_f$ versus flow or Reynolds number; $f$ versus Reynolds number for nine pipes with correlations; and fitting $K$ versus Reynolds number/equivalent length. Use correct water-flow units, condition identifiers, captions and physically plausible values. | _____ |
| Discussion                 | 15         | Quantify calibration and $C_f$ results. Explain flow-regime, diameter, material and roughness effects on $f$ and fitting effects on $K$ . Compare correlations, address anomalies and model limitations, and answer all objectives.                                                                                       | _____ |
| References                 | 10         | Cite relevant correlations and sources for technical claims. Provide complete entries and cite every listed source in the text. Meet the source-count requirement when assigned.                                                                                                                                          | _____ |
| Appendix I: Data/Graphs    | 10         | Provide complete raw and processed calibration, orifice, nine-pipe and fitting data. Include raw calibration records, condition identifiers, units, readable tables/plots, and replicates when collected or assigned. Make results traceable to measurements.                                                             | _____ |
| Appendix II: Error         | 10         | Quantify volume/time, pressure-instrument, temperature/viscosity and calibration uncertainty. Propagate uncertainty or test sensitivity in $C_f$ , $f$ and $K$ . Distinguish systematic and random error, assess effects on conclusions, and propose realistic improvements.                                              | _____ |
| Appendix III: Calculations | 10         | Identify a trial. Show water-flow conversion, Reynolds number, $C_f$ , friction factor/head loss, $K$ and equivalent length where used. Include equations, substitutions, units and results traceable to reported values.                                                                                                 | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                                           | _____ |

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# Humidification

## Final Report Rubric • 100 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                                        | Score |
|----------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | State the unit, objective and actual air-flow, inlet-temperature, water-flow and pressure conditions. Report numerical $k_y$ and $h$ results, correlation comparisons, main trends and limitations. Include NTU/HTU results if reported or assigned.                                                                    | _____ |
| Introduction               | 10         | Describe the purpose, applications and countercurrent wetted-wall process and apparatus. State varied and controlled variables, measured outputs and analyses. Support background information with relevant citations.                                                                                                  | _____ |
| Theory                     | 10         | Present heat/mass flux laws, humidity ratio, balances for $k_y/h$ , $Re$ , $Sc$ , $Sh$ and $Sh = 0.023 Re^{0.83} Sc^{0.44}$ . Define quantities, units and assumptions; cite sources. Include NTU/HTU and the Lewis relation $h/(k_y C_s) \approx 1$ if reported or assigned.                                           | _____ |
| Results                    | 15         | Report temperatures, relative humidity and coefficients; plot $k_y$ and $h$ versus $Re$ and $\log_{10}(Sh/Sc^{0.44})$ versus $\log_{10}(Re)$ ; compare the empirical correlation. Include NTU versus inlet temperature by airflow, HTU and Lewis results if reported or assigned. Label axes, units and conditions.     | _____ |
| Discussion                 | 15         | Quantitatively explain airflow/ $Re$ and inlet-temperature effects on $k_y$ , $h$ and relative humidity using residence time and driving force. Compare correlations and address assumptions, deviations and improvements. Discuss NTU/HTU and Lewis comparisons if reported or assigned.                               | _____ |
| References                 | 10         | Meet the assigned source-count requirement. Use credible sources, cite outside claims in the text, provide complete entries and cite every listed source in the report.                                                                                                                                                 | _____ |
| Appendix I: Data/Graphs    | 10         | Provide trial-indexed air/water flows, pressure and steam conditions, inlet/outlet temperature and humidity, and wall/interface/reservoir temperatures. Include humidity, $G_s$ , $k_y$ , $h$ , $Re$ , $Sc$ and $Sh$ calculations, units and supplemental plots. Include NTU/HTU/Lewis results if reported or assigned. | _____ |
| Appendix II: Error         | 10         | Identify specific error sources and quantify replicate or instrument uncertainty, or percent difference. Relate errors to humidity, transfer coefficients and correlation comparisons. Propose realistic improvements.                                                                                                  | _____ |
| Appendix III: Calculations | 10         | Identify one trial. Show properties and sources, humidity ratios, $G_s$ , $k_y$ , $h$ , $Re$ , $Sc$ , $Sh$ and $\log_{10}(Sh/Sc^{0.44})$ . Include NTU/HTU and the Lewis test if reported or assigned. Show substitutions, units and results matching the report.                                                       | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                                         | _____ |

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After the next lab period: 0 score.

Content score: \_\_\_\_\_ / 100    Late multiplier: 1 / 0.5 / 0    Final score: \_\_\_\_\_ / 100

# Membrane Air Separation

## Final Report Rubric • 100 points

Team: \_\_\_\_\_ Section: \_\_\_\_\_ Grader: \_\_\_\_\_ Date: \_\_\_\_\_

Award partial credit for incomplete or incorrect work. Score each section in 0.5-point increments.

| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                                                                         | Score |
|----------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | State the objective, two-module method, pressure/flow ranges and configurations. Report numerical KO <sub>2</sub> and KN <sub>2</sub> with units, selectivity and recovery, the best operating condition and a supported series/parallel comparison.                                                                                                     | _____ |
| Introduction               | 10         | Describe membrane-separation significance, applications, solution diffusion and the actual Prism apparatus. State objectives, variables and the series/parallel scope. Cite relevant technical background and outside claims.                                                                                                                            | _____ |
| Theory                     | 10         | Present Fick/Henry relationships, solution diffusion, permeability, molar flow/flux, overall K, absolute partial pressures, separation factors, O <sub>2</sub> /N <sub>2</sub> recovery and balances. Distinguish overall K from area-normalized permeance and permeability. Define units and cite equations; include stage cut if reported or assigned. | _____ |
| Results                    | 15         | Show conversions and balance checks; KO <sub>2</sub> /KN <sub>2</sub> versus pressure by flow and configuration; required separation factors and recoveries; series/parallel comparison and optimum. State complete conditions and coefficient bases, with units and readable tables/figures.                                                            | _____ |
| Discussion                 | 15         | Explain pressure, flow and configuration effects using transport mechanisms and data. Evaluate purity, recovery and throughput tradeoffs and the optimum. Compare literature; address balance closure, temperature, anomalies and the module-area basis.                                                                                                 | _____ |
| References                 | 10         | Use relevant, credible sources and cite technical claims in the text. Provide complete entries and cite every listed source. Meet the source-count requirement when assigned.                                                                                                                                                                            | _____ |
| Appendix I: Data/Graphs    | 10         | Provide the complete run matrix: raw tube/shell flow, oxygen composition, pressure, temperature, configuration and module. Include processed K, separation, recovery and balances, with units, supporting plots and traceability from measurements to results.                                                                                           | _____ |
| Appendix II: Error         | 10         | Quantify applicable oxygen-sensor, flow-meter, pressure, temperature and steady-state errors using sensitivity or uncertainty propagation. Assess effects on K, separation, recovery and conclusions. Propose realistic corrections and improvements.                                                                                                    | _____ |
| Appendix III: Calculations | 10         | Identify a run. Convert measured volumetric flow to molar flow using stated reference conditions. Show gauge-to-absolute and species partial pressures, species permeation, both K values, balances, separation factors and recoveries. Include substitutions and consistent units.                                                                      | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                                                                          | _____ |

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# Non-Newtonian Fluid

## Final Report Rubric • 100 points

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| Section                    | Points     | Grading criteria                                                                                                                                                                                                                                                                                    | Score |
|----------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Executive Summary          | 10         | State the objective, method and operating conditions. Give numerical m and n values, units and ranges, fluid classification, geometry/shear-rate trends and a conclusion supported by the results.                                                                                                  | _____ |
| Introduction               | 10         | Describe motivation, applications and the Carbopol/capillary system. State the objective, scope, variables and experimental design. Present an organized introduction with citations supporting relevant background information.                                                                    | _____ |
| Theory                     | 10         | Present Newtonian and power-law behavior; m/n meanings, units and classification; capillary logarithmic relationships and slope/intercept fitting; wall shear stress and velocity-gradient/shear-rate relationships. Define assumptions and cite equations. Include apparent viscosity if reported. | _____ |
| Results                    | 15         | Present traceable tube, pressure and Q data; logarithmic plots/fits; m and n with units; shear stress versus velocity gradient/shear rate; quantitative literature comparison; and L/D comparison. Use clear captions, units, appropriate significant figures and useful statistics.                | _____ |
| Discussion                 | 15         | Support fluid classification with n and measured data. Interpret m/n and geometry effects, compare Carbopol literature quantitatively, address anomalies and model limitations, and answer the objectives with evidence.                                                                            | _____ |
| References                 | 10         | Use relevant sources and cite outside claims in the text. Provide complete, consistent entries and cite every listed source. Meet the source-count requirement when assigned.                                                                                                                       | _____ |
| Appendix I: Data/Graphs    | 10         | Provide all tube and pressure data with units, replicates when collected, and supporting tables/plots. Make m/n values and plots traceable to raw measurements. Use readable captions and appropriate significant figures.                                                                          | _____ |
| Appendix II: Error         | 10         | Identify random and systematic errors. Evaluate repeatability when replicates exist; otherwise quantify uncertainty or sensitivity. Quantify effects on m, n or classification, relate them to conclusions, and propose realistic improvements.                                                     | _____ |
| Appendix III: Calculations | 10         | Identify a trial. Convert timed volume or timed mass/density to Q according to the measurements. Show transformed plot variables, m/n from slope/intercept, wall shear stress and velocity gradient/shear rate. Include substitutions and intermediate units; apparent viscosity if reported.       | _____ |
| <b>Content total</b>       | <b>100</b> |                                                                                                                                                                                                                                                                                                     | _____ |

### Late-submission policy

By the due date: full earned score.

After the due date and up to the next lab period: 50% of the earned score.

After the next lab period: 0 score.

Content score: \_\_\_\_\_ / 100    Late multiplier: 1 / 0.5 / 0    Final score: \_\_\_\_\_ / 100